## Data processing

Check sequencing data quality
*FastQC*
Output: HTML report

Adapter trimming
***Skewer*** ***Cutadapt***
***TRIMMOMATIC***
Output: trimmed FastQ

Alignment (and pre-alignments)
***bowtie2*** ***bwa***
Output: BAM

Duplicate read removal
***Picard*** ***samblaster***
Output: deduplicated BAM

Tn5 offset adjustment
***deepTools*** ***Custom scripts***
Output: offset-adjusted BAM

## Peak calling merging and counting

Peak calling
***MACS2*** ***Genrich***
Output: BED peak regions and summits

Peak merging
***bedtools*** ***Custom scripts***
Output: BED

Count transpositions in peaks
***bedtools*** ***Custom scripts***
Output: insertion counts matrix

## Downstream analysis

Differential accessibility
***DESeq2*** ***edgeR*** ***limma***
Considerations:
- Normalization
- Batch effect correction
- Sensitivity versus specificity

Motif enrichment
***Homer*** ***MEME-AME***
Considerations:
- Background peak set
- Motif set and origin
- False positive TFBSs

TF footprinting
***HINT-ATAC*** ***BaGFoot***
***TOBIAS***
Considerations:
- Tn5 bias
- Tn5 offset
- De novo versus motif-centric
- Supervised versus unsupervised
- Single-site versus meta

Track visualization
***IGV*** ***WashU EpiGenome Browser***
***UCSC Browser*** ***Custom scripts***
Considerations:
- Normalization
- Binning

Integrative multi-omic analysis
***Custom scripts***
Frequently paired data types:
- RNA-seq
- ChIP-seq / CUT&RUN
- HiC and HiChIP
- DNA methylation